A finite Cantor subgroup has a different smoothness fraction

On the complete reduced-Mumford polynomial-pair model for $y^2 = x^5 - x$ over $\mathbf{F}_3$, Lean proves all 64 Cantor additions and unique reduced output. The proper cyclic subgroup generated by $(u, v) = (X, 0)$ has order two. It is entirely smooth, while seven of the eight ambient representations are smooth. [pcc:unverified]

Exact sampling quantity	Ambient model	Cyclic subgroup
Smooth, including identity	$7/8$	$2/2 = 1$
Nonzero smooth numerator, whole-group denominator	$6/8 = 3/4$	$1/2$

Lean disproves the assertion that every cyclic subgroup has the ambient smoothness fraction in this finite arithmetic model. Formal identification with the geometric Picard/Jacobian group remains open. Fixed order two over an odd field does not refute a qualified binary large-prime, typical-subgroup or asymptotic heuristic. [pcc:unverified]

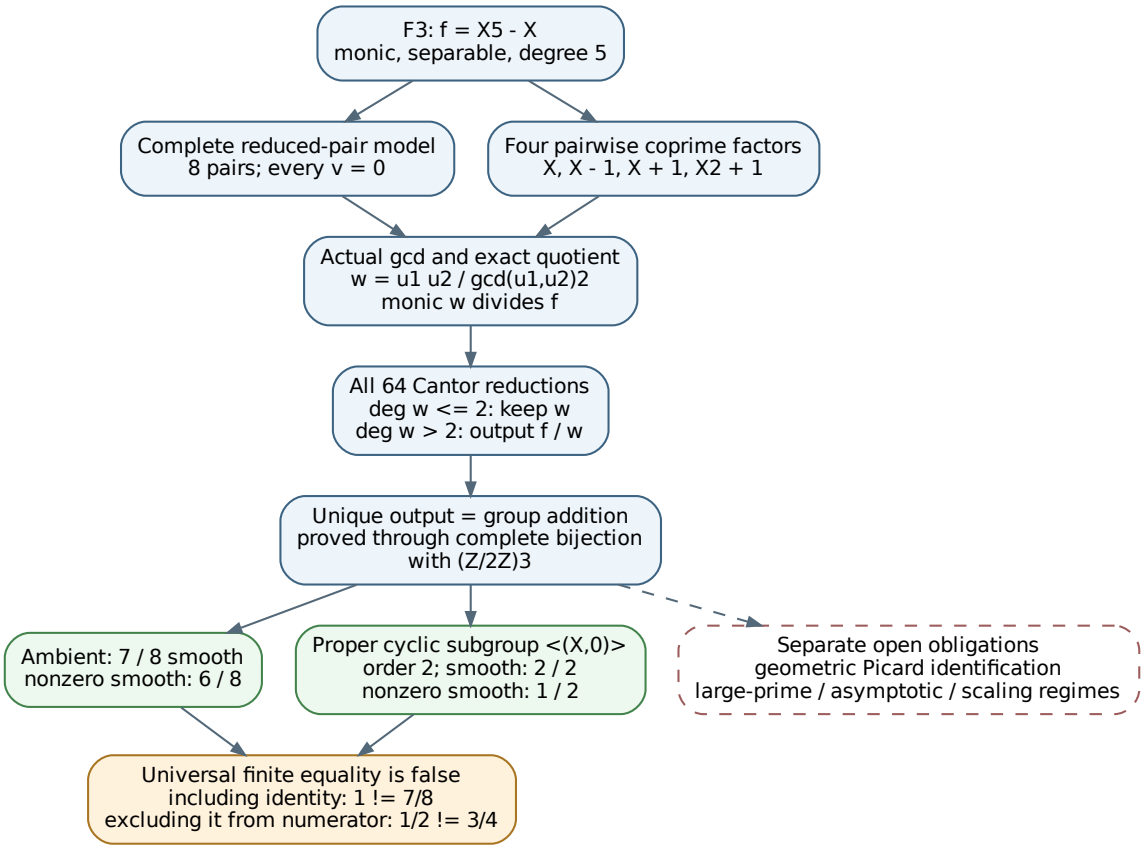


Figure 1: Actual polynomial arithmetic precedes the group and subgroup comparison. Dashed obligations remain open.

Proof, sources and verification

The base model ranges over every actual pair with u monic, $\deg u \le 2$, $\deg v < \deg u$ and $u \mid v^2 - f$. Its complete coefficient classification gives eight pairs; all have $v = 0$. The curve polynomial is proved monic, degree five and squarefree. [pcc:unverified]

Lean proves monicity and pairwise coprimality of $X, X-1, X+1, X^2+1$, using explicit Bezout certificates. For every input, the actual normalized gcd d and monic composition polynomial w satisfy $d^2 w = u_1 u_2$ and $w \mid f$. The full gcd with $v_1 + v_2$ is d, and explicit standard-composition witnesses give zero for the v numerator. Reduction keeps w when its degree is at most two, and otherwise proves $w\,u = f$. Exact monic-product degrees and a kernel Boolean certificate select all 64 branches. [pcc:unverified]

Output uniqueness and an explicit bijection with $(\mathbf{Z}/2\mathbf{Z})^3$ establish equivalence between this Cantor relation and group addition. The cyclic subgroup is proved proper, with exact coordinates, order, cardinality and splitting counts. Transporting an arbitrary group onto eight labels alone would not establish these statements. [pcc:unverified]

Proof obligation	Module / declaration
Gcd, composition, all reductions	MumfordCantor / cantor_all_pairs
Unique actual output	MumfordCantor / cantor_output_unique
Group matches the Cantor relation	MumfordCantorGroup / cantor_iff_addition
Direct universal finite disproof	MumfordCantorGroup / universal_finite_cyclic_equality_false
Nonzero counts and ratios	MumfordCantorGroup / useful_smooth_fractions

The combined binding is:

`MumfordCantorGroupAudit.cantor_subgroup_audit_certificate`.

The whole tracked project gate passes 170 declarations, 0 failures and 0 trusted-native exceptions. Only propext, Classical.choice and Quot.sound are accepted. Lean 4.35.0 and Mathlib revision `56a005855ea17e0b0958f73ec801ae140b0a1679` are pinned. Earlier failed builds remain retained; one raw diagnostic log is losslessly compressed with raw/archive digests. Lake retains module-system advisories; no warning-free build is claimed. [pcc:unverified]

Source commit `5ffeedb4` precedes packet `PCC-20261109-4a0f82` and the scalar supplement. The packet's full local check passes at 20:56:49Z on producer host `762f5fc120c9`. Signed standing remains unverified. The scalar supplement is `IDC1h95637a8da70e67cd`. The supplement accepts 32 derived and 11 extra seeded points over $2^{61} - 1$ for cancellation/quintic-factorization identities only: five variables, degree bound six. It does not prove gcd selection, the group law or counts. Those arithmetic statements rest on Lean. [pcc:unverified]

Read Stoll (2014), Definition 2.2 (p. 5), Lemmas 4.16–4.17 and Theorem 4.18 (pp. 23–25), including proofs, and Lemma 5.2 (p. 30). PDF page 24 was visually inspected. These expose the characteristic-not-two, squarefree and geometric representation conditions and actual arithmetic rules. Cantor1987's publisher PDF remains unread after 403. Gaudry's fixed-genus limit and MMT's binary large-prime heuristic are preserved. No new native curve population, paid compute or independent-host validation ran here. [pcc:unverified]

Exact hashes, source URLs, proof map and reproduction commands are in ../../CANTOR_SUBGROUP.md, ../../cantor-sources.json and ../../cantor-verification.json. The original fourteen-item audit remains partial: geometric identification, native multi-field/scaling work and qualified large-prime/asymptotic obligations remain open. No performance or curve-comparison metric changed; canonical scoreboards are unchanged. [pcc:unverified]